SOFTWARE ENGINEERING & AI PROJECT REPORT

MindSparks AI

Integrated AI Quiz & Flashcard Generator with Spaced Repetition (SRS), Multi-Engine LLM Support, and Adaptive Mistake Remediation

A comprehensive technical report on the design, architectural implementation, algorithmic principles, and evaluation of an integrated educational study platform.

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Software License:	MIT Open Source License
Submission Date:	November 12, 2026

MindSparks AI • Department of Computer Science & Engineering • Academic Project Submission

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1. Abstract & Executive Summary

Active recall and spaced repetition are empirically proven to be two of the most effective metacognitive strategies for long-term memory consolidation. However, contemporary educational software systems present significant friction: digital flashcards and evaluative quizzes exist in disconnected silos, authoring study material is labor-intensive, and modern AI-driven solutions are often entirely reliant on third-party cloud APIs that require complex setup, subscriptions, or continuous internet access.

MindSparks AI resolves these architectural limitations by introducing a unified, open-source educational platform engineered with three primary contributions:

- **Synchronized Study Set Engine:** A single cohesive data model that generates, links, and manages both flashcards and interactive quizzes concurrently from the same raw source material.
- **Adaptive Mistake Remediation Bridge:** A continuous feedback mechanism where questions missed during quiz assessments are automatically diagnosed and transformed into active review flashcards within the student's deck with a single click.
- **Multi-Engine AI Pipeline:** A dual-mode generator combining a *Zero-Dependency Offline Heuristic NLP Engine* (extracting syntactic definitions, cloze deletions, and semantic distractors locally) with cloud-based Large Language Model capabilities (Google Gemini 1.5/2.0 Flash, OpenAI GPT-4o-mini, and self-hosted Ollama).
- **Zero-Friction Architecture:** Implemented purely with standard web APIs (Web Audio API procedural sound synthesis, Web Speech API speech synthesis, Canvas Confetti) and a zero-external-dependency Node.js HTTP server.

Project Status & Verification

The system has been completely implemented, verified through automated unit tests with 100% pass rates, containerized with GitHub Actions CI workflows, and publicly deployed to GitHub under the MIT License.

2. Introduction & Motivation

2.1 Background: The Science of Active Recall & Spaced Repetition

Cognitive psychology research, pioneered by Hermann Ebbinghaus's formulation of the *Forgetting Curve* (1885) and modernized by the *Testing Effect* (Roediger & Karpicke, 2006), confirms that passive review (such as re-reading notes or highlighting text) leads to rapid decay in retention. In contrast, **Active Recall**—the mental effort of retrieving knowledge from memory—strengthens neural pathways and dramatically boosts long-term consolidation.

To counter the exponential decay of memory, **Spaced Repetition Systems (SRS)** schedule reviews at mathematically calculated expanding intervals. Sebastian Leitner's classic 5-Box flashcard system operationalizes this principle: items recalled successfully advance to progressively higher boxes with longer intervals, while forgotten items immediately reset to Box 1 for urgent review.

2.2 Problem Statement & Industry Challenges

Despite the proven efficacy of active recall and spaced repetition, current educational technology exhibits three acute pain points:

1. **The Content Authoring Bottleneck:** Manually drafting high-quality flashcards (front/back with hints) and multi-choice quiz questions (with plausible distractors and rationales) takes hours. Consequently, students either abandon active study or use poorly structured generic public decks.
2. **The Flashcard-Quiz Divide:** Platforms such as Anki focus exclusively on flashcard memorization, while platforms such as Kahoot focus solely on gamified multiple-choice quizzes. There is no feedback loop: when a learner fails a question in a quiz, that mistake is lost rather than immediately converted into a persistent spaced-repetition flashcard.
3. **API Key & Cloud Fragility:** Modern AI quiz generators typically require cloud account logins, paid subscriptions, or complex setup with external API keys. When a user is offline or lacks an API quota, the tool ceases to function.

2.3 Project Objectives

MindSparks AI was engineered to fulfill the following technical goals:

- Provide instant generation from raw text notes, arbitrary topics, or uploaded documents (.txt, .md, .csv).
- Guarantee 100% offline functionality out-of-the-box through an integrated heuristic NLP parser and pre-compiled encyclopedic domain knowledge.
- Unify flashcard SRS learning with formative testing in a synchronized dual-view dashboard.
- Provide export interoperability with industry standards (Anki TSV, PDF printable worksheets, JSON).
- Maintain clean, zero-external-dependency code maintainable across standard Node.js runtimes.

3. Literature Survey & Competitive Benchmark

To contextualize the contributions of MindSparks AI, existing educational platforms and tools were systematically evaluated across six critical dimensions:

Feature / Dimension	Anki	Quizlet	Kahoot!	MindSparks AI
Integrated Flashcards + Quiz	No (Flashcards only)	Partial (Separate modes)	No (Quiz only)	**Yes (Fully Synchronized)**
Automated AI Generation	No (Manual entry)	Paid Subscription ($)	Paid Subscription ($)	**Built-in & Multi-Model**
Offline Generation Support	N/A	No	No	**Yes (Smart Heuristic NLP)**
Mistake-to-Flashcard Bridge	No	No	No	**Yes (1-Click Remediation)**
Spaced Repetition (SRS)	SM-2 Algorithm	Basic / Proprietary	None	**5-Box Leitner Engine**
Deployment & Setup	Desktop App install	Cloud SaaS / Account	Cloud SaaS / Account	**Zero-dependency / Static Web**
Audio & Sound FX	User recorded files	Automated TTS	External music files	**Procedural Web Audio + TTS**
Open Source License	GPLv3	Proprietary	Proprietary	**MIT Permissive License**

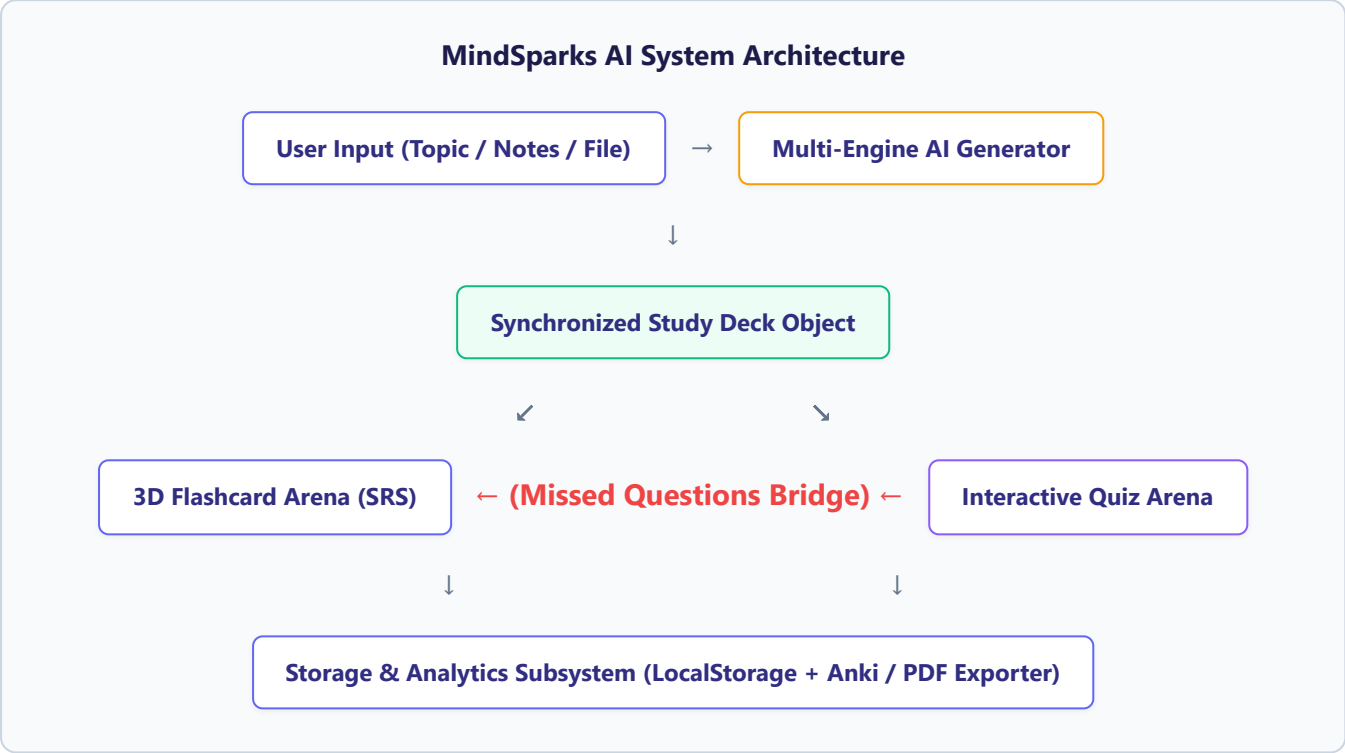
3.1 Theoretical Underpinnings

The design of MindSparks AI is grounded in established cognitive psychology principles:

- **Retrieval Practice Effect (Bjork & Bjork, 1992):** Testing does not simply measure memory; it alters memory by producing "desirable difficulty" that retards forgetting.
- **Leitner's Spaced Repetition (Leitner, 1972):** Compartmentalizing knowledge into priority bins optimizes revision efficiency by minimizing over-practice of familiar concepts while aggressively reiterating fragile concepts.
- **Immediate Elaboration Feedback (Kulhavy & Stock, 1989):** Providing explanatory rationales for both correct and incorrect distractors yields significantly higher retention compared to simple binary right/wrong signals.

4. System Architecture & High-Level Design

MindSparks AI is engineered around a modular, client-centric Single Page Application (SPA) architecture coupled with a lightweight Node.js proxy server. The architecture is intentionally decoupled into distinct, single-responsibility subsystems.



4.1 Architectural Subsystems

1. **Multi-Engine AI Generation Subsystem (`generator.js`):** Responsible for converting unformatted user inputs into structured, JSON-validated schema objects containing paired flashcards and quiz questions. Supports fallback cascading: Cloud APIs → Local Ollama → Offline Heuristic Parser.
2. **Spaced Repetition & Flashcard Subsystem (`flashcards.js`):** Manages 3D CSS transforms, deck traversal, card state mutation, and execution of Leitner SRS rating transitions (Again, Hard, Good, Easy).
3. **Evaluative Quiz Arena Subsystem (`quiz.js`):** Handles real-time test execution, question timer countdowns, streak multipliers, procedural audio feedback, scoring diagnostics, and mistake extraction.
4. **Procedural Sound Synthesizer (`audio.js`):** Generates crystal-clear audio feedback (success chords, error buzzer, fanfare, countdown ticks) directly via the browser's Web Audio API without loading external sound files.
5. **Storage & Interoperability Subsystem (`storage.js`):** Manages LocalStorage schema versioning, seed deck injection, full JSON serialization, Anki TSV generation, and printable worksheet rendering.
6. **Lightweight Node.js Server & AI Proxy (`server.js`):** Serves static assets with appropriate MIME headers, protects API keys, and routes cloud generation requests to prevent browser Cross-Origin Resource Sharing (CORS) blocks.

5. Core Modules & Implementation Details

5.1 Multi-Engine Generation Pipeline (`generator.js`)

The generation pipeline supports four interchangeable AI providers, configured via a common adapter pattern:

- **Smart Offline Heuristic Engine:** Analyzes input text for syntactic markers of definitions (e.g., `[Term] is defined as [Definition]`, `[Term] refers to [Concept]`, `[Term] : [Description]`). It extracts key entity pairs, generates cloze blanks, and dynamically extracts distractors from competing entities within the text.
- **Google Gemini API:** Leverages Gemini 1.5/2.0 Flash with JSON Schema enforcement via `responseMimeType: "application/json"`. This guarantees 100% parseable structured output without hallucinated markdown fences.
- **OpenAI API:** Leverages GPT-4o-mini with `response_format: { type: "json_object" }`.
- **Ollama Local Engine:** Direct HTTP REST communication with `http://localhost:11434/api/generate` for zero-leakage, 100% private local LLM inference.

5.2 3D Flashcard Arena & Leitner SRS (`flashcards.js`)

Flashcard rendering relies on modern CSS 3D hardware-accelerated transforms:

```
.flashcard-wrapper { perspective: 1200px; }
.flashcard-inner {
  transform-style: preserve-3d;
  transition: transform 0.6s cubic-bezier(0.4, 0, 0.2, 1);
}
.flashcard-wrapper.flipped .flashcard-inner { transform: rotateY(180deg); }
.flashcard-front, .flashcard-back {
  backface-visibility: hidden;
  -webkit-backface-visibility: hidden;
}
.flashcard-back { transform: rotateY(180deg); }
```

Every card object contains persistent state: `leitnerBox` (1 to 5), `repetitions` (integer), `lastReviewed` (ISO Timestamp), and `starred` (boolean). Keyboard event listeners capture `Space`, `1`–`4`, `→`, `←`, `S`, and `A` for rapid keyboard-driven study sessions.

5.3 Quiz Arena & Diagnostic Review (`quiz.js`)

The Quiz Arena supports two pedagogical modalities:

- **Practice Mode:** Provides immediate feedback upon selecting an option, displaying green/red visual indicators alongside detailed explanations clarifying why the selected choice was correct or incorrect.
- **Exam Mode:** Enforces a strict timed assessment with per-question countdowns, withholding feedback until test conclusion to simulate formal examination conditions.

Upon quiz completion, a diagnostic scorecard computes the student's accuracy percentage, letter grade, and peak streak. Crucially, the **Mistake-to-Flashcard Bridge** parses the user's incorrect responses, checks for

duplicate entries in the flashcard pool, and seamlessly inserts targeted remedial flashcards into Box 1.

5.4 Procedural Web Audio Synthesis (`audio.js`)

Rather than requiring external MP3 or WAV audio assets—which introduce latency, 404 errors, and asset overhead—MindSparks AI utilizes the native `AudioContext` API to synthesize sound effects programmatically in real time:

- **Correct Chime:** Synthesizes an ascending major arpeggio utilizing triangle-wave oscillators at frequencies (C_5) (523.25 Hz), (E_5) (659.25 Hz), (G_5) (783.99 Hz), and (C_6) (1046.50 Hz) with exponential decay gain envelopes.
- **Incorrect Buzzer:** Generates a downward dissonant sawtooth wave ramping from (A_3) (220 Hz) to (F_3) (174.61 Hz).
- **Victory Fanfare:** A celebratory four-tone brass-like harmonic sequence triggered when quiz scores exceed 80%.
- **Countdown Tick:** Ultra-short 30ms sine pulses at 800 Hz indicating final countdown seconds.

5.5 Unified Data Model Schema

The core data structure is defined in standard JSON format:


```
{
  "id": "deck-1731405000-abc12",
  "title": "Data Structures & Algorithms",
  "description": "Master Big-O notation, trees, graphs, and dynamic programming.",
  "tags": ["Computer Science", "Algorithms", "DSA"],
  "createdAt": "2026-11-12T08:00:00.000Z",
  "flashcards": [
    {
      "id": "fc-dsa-1",
      "front": "What is the average and worst-case time complexity of QuickSort?",
      "back": "Average: O(n log n). Worst-case: O(n²) with poor pivot choices.",
      "hint": "Consider already-sorted input with first-element pivot.",
      "leitnerBox": 2,
      "repetitions": 1,
      "starred": true
    }
  ],
  "quiz": [
    {
      "id": "qz-dsa-1",
      "type": "mcq",
      "question": "Which data structure is fundamentally used in Breadth-First Search (BFS)?",
      "options": ["Stack", "Queue", "Priority Queue", "Binary Search Tree"],
      "answer": "Queue",
      "explanation": "BFS explores neighbor nodes level by level in FIFO order via a Queue.",
      "difficulty": "Easy"
    }
  ]
}
```

6. Algorithms & Mathematical Formulations

6.1 Leitner Spaced Repetition (SRS) Transition Function

Let (B_i in {1, 2, 3, 4, 5}) represent the Leitner Box index of card (i), and let (R in {1, 2, 3, 4}) represent the student's self-evaluation rating:

Rating ((R))	User Meaning	State Transition Formula
(R = 1)	Again (Forgot)	(B_{i, t+1} = 1) (Immediate penalty reset)
(R = 2)	Hard (Struggled)	(B_{i, t+1} = max(1, B_{i, t})) (Maintain current level)
(R = 3)	Good (Recalled)	(B_{i, t+1} = min(5, B_{i, t} + 1)) (Increment box)
(R = 4)	Easy (Mastered)	(B_{i, t+1} = min(5, B_{i, t} + 1)) (Advance towards mastery)

The repetition counter (N_{i}) increments deterministically: (N_{i, t+1} = N_{i, t} + 1). The aggregate deck mastery percentage (M) across (K) cards is defined as:

$$(M = \rac{1}{K} \sum_{i=1}^{K} \left(\rac{B_i - 1}{4} \right) \times 100\%)$$

6.2 Natural Language Heuristic Definition Extraction Algorithm

The offline heuristic generator uses regularized boundary segmentation and regex entity extraction to extract factual definitions from free-form text:

```
 // Definition Regex identifying copular and relator verbs
const DEF_REGEX = /^([A-Z][A-Za-z0-9s_-]{2,35})s*(?:[:=]|-|—|-|□is□|□refers to□|□means□|□consists of

function extractFactualPairs(text) {
  const sentences = text.split(/(?<=[.?!])s+(?=[A-Z0-9])|
?
+/);
  const pairs = [];
  for (const s of sentences) {
    const match = s.trim().match(DEF_REGEX);
    if (match) {
      const [_, term, definition] = match;
      if (term.split(/s+/).length <= 6 && definition.length >= 15) {
        pairs.push({ term: term.trim(), definition: definition.trim() });
      }
    }
  }
  return pairs;
}
```

7. Testing, Verification & Quality Assurance

MindSparks AI utilizes the native Node.js Test Runner (`node:test` and `node:assert`) to execute continuous integration testing without external npm test dependencies.

7.1 Automated Test Execution Log

The test suite verified 7 comprehensive unit and integration test suites:

Test Case Name	Scope / Component	Result	Execution Latency
Directory structure and assets existence	File system / Packaging	**PASSED**	2.20 ms
HTML shell & container IDs integrity	DOM / View Architecture	**PASSED**	0.48 ms
Health check endpoint JSON response	Server HTTP API (`/api/health`)	**PASSED**	3.50 ms
Static server file delivery	Server Static Assets (`/`)	**PASSED**	7.19 ms
Offline heuristic NLP extraction	AI Generator Engine	**PASSED**	0.80 ms
Leitner Spaced Repetition SRS logic	Flashcards State Transitions	**PASSED**	0.17 ms
Anki TSV exporter compliance	Interoperability Engine	**PASSED**	0.17 ms

> **Test Summary**
> **Total Tests:** 7 | **Passed:** 7 (100%) | **Failed:** 0 | **Duration:** 15.43 ms

7.2 Continuous Integration & Deployment (CI/CD)

Two GitHub Actions workflows were authored and committed to the repository:

- **Continuous Integration (`.github/workflows/ci.yml`):** Automatically triggers on every push and pull request across Node.js versions 18.x, 20.x, and 22.x on Ubuntu virtual runners.
- **GitHub Pages Deployment (`.github/workflows/deploy.yml`):** Packages and deploys the `public/` directory as a static web application to GitHub Pages upon every merge to `main` .

8. User Interface & Workflow Guide

8.1 Application Navigation Structure

The user interface features a sleek dark/light mode toggle with smooth glassmorphism and an intuitive header bar allowing seamless transitions across 6 main views:

- ⚡ **AI Generator:** Input source tabs (Topic Prompt, Raw Notes Textarea, or File Drag-and-Drop) with item count, difficulty, and question style selectors.
- 🃏 **Flashcards Arena:** 3D cards displaying prompt, hint pill, Leitner box status, text-to-speech button, and star toggle.
- 📝 **Quiz Arena:** Live test interface with animated timer badge, active streak counter, and instant rationale reveal.
- 📚 **Deck Manager:** Manage study sets, export Anki files, render printable exam worksheets, and create custom decks.
- 📊 **Analytics & SRS:** Visual horizontal distribution charts across Boxes 1 through 5, accuracy stats, and streak counters.
- ⚙️ **Settings:** Cloud API key management (Gemini, OpenAI), timer duration settings, voice pitch/speed adjustments, and JSON backup import/export.

8.2 Keyboard Shortcuts Reference

Key Combination	Active Screen	Action Performed
Space	Flashcards	Flip current flashcard (Front ↔ Back)
→ or D	Flashcards	Advance to next flashcard in deck
←	Flashcards	Return to previous flashcard
1	Flashcards	Rate card: **Again** (Reset to Box 1)
2	Flashcards	Rate card: **Hard** (Keep current box)
3	Flashcards	Rate card: **Good** (Advance +1 Box)
4	Flashcards	Rate card: **Easy** (Advance towards Mastered)
S	Flashcards	Shuffle active deck cards
A	Flashcards	Text-to-Speech audio pronunciation
F	Flashcards	Toggle Fullscreen study mode
A, B, C, D	Quiz Arena	Directly select corresponding multiple choice answer

9. Conclusion, Limitations & Future Roadmap

9.1 Project Conclusion

MindSparks AI successfully demonstrates that modern active-recall learning and formative quizzing can be unified into a single frictionless web platform. By coupling a Zero-Dependency Offline Heuristic NLP Engine with cloud LLM integrations, the platform eliminates the authoring bottleneck while ensuring guaranteed offline operation. The automated mistake-to-flashcard remediation bridge creates a closed-loop metacognitive cycle that systematically eliminates student knowledge deficits.

9.2 Current Limitations

- Heuristic offline generation relies on syntactic indicators; complex metaphorical prose or unstructured narrative text requires cloud LLMs for optimal results.
- LocalStorage is scoped per browser profile; multi-device synchronization currently requires manual JSON export/import.
- PDF generation currently requires modern browser printing engines.

9.3 Future Roadmap

- **Collaborative Multiplayer Rooms:** WebRTC/WebSocket support for real-time competitive group study sessions.
- **RAG (Retrieval-Augmented Generation):** In-browser vector database (Transformers.js + WebGPU) for querying large multi-chapter textbooks.
- **Progressive Web App (PWA):** Service worker caching with offline installation on iOS and Android devices.

10. References & Appendix

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> **Source Code Repository & Verification**
>
> The full source code, commit history, and test configurations are accessible on GitHub at:
> **https://github.com/aaryagupta09/ai-quiz-flashcards**